

**EXPERIMENTS AND MODELING OF THE HEAT TRANSFER OF IN-LINE
SQUARE PIN FIN HEAT SINKS WITH TOP BY-PASS FLOW**

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ABSTRACT

This paper presents experimental results on the heat transfer characteristics of in-line square pin fin heat sinks with and without top by-pass. A self-consistent set of aluminum heat sinks were utilized, where the pin height was varied from 12.5 mm to 22.5 mm, the pin pitch was varied from 3.4 mm to 5.8 mm and the base dimensions were kept fixed at 25 x 25 mm. The overall base to ambient thermal resistance was measured as a function of Reynolds number and by-pass height. Experimental results were then compared with predictions based on a simple one-dimensional "two-branch by-pass model". Comparisons were made with the data using heat transfer coefficients available in the literature for infinitely long tube bundles. It was shown that there is a good agreement between the temperature predictions based on the model and the experimental data at high approach velocities for tall heat sinks, however the discrepancy between the computations and experiments increases as the approach velocity and heat sink height decrease. The validated model was used to identify optimum pin spacing as a function of clearance ratio.

Key words: pin fins, heat sink, in-line arrangement, by-pass, heat transfer, two-branch by-pass model, experimental study.

NOMENCLATURE

A_b	base surface area of a given control volume (m^2)
A_{cs}	cross-sectional area of a single fin (m^2)
A_f	area of a single fin exposed to heat transfer (m^2)
A_w	wetted heat sink area (m^2)
a	fin side (m)
b	distance between two fins in the flow direction (m)
C	constant defined in Eq. (13)
CL	clearance ratio (L/H)
c_p	specific heat ($J/(kg \cdot K)$)
d_f	fin (hydraulic) diameter (m)
H	height of the finned section (fin height) (m)
h	heat transfer coefficient ($W/(m^2 \cdot K)$)
k	thermal conductivity of fluid ($W/(m \cdot K)$)
k_f	thermal conductivity of fin material ($W/(m \cdot K)$)
L	height of the top by-pass section (top clearance) (m)
L_c	effective fin length (m)
C	constant defined in Eq. (13)
Nu	Nusselt number based on d_f
P_L	non-dimensional longitudinal pitch (S_L/d_f)
P_T	non-dimensional transverse pitch (S_T/d_f)
Pr	Prandtl number at mean fluid temperature
Pr_s	Prandtl number at surface temperature
q_{conv}	rate of heat transfer convected through heat sink (W)

q_{lss}	total amount of heat loss (W)
q_{tot}	rate of total power applied to heat sink (W)
R_{th}	thermal resistance (K/W)
$(R_{th})_0$	zero clearance thermal resistance (K/W)
Re_{av}	Reynolds number based on the average velocity along the fin surface and the fin diameter
S_L	longitudinal pitch (m)
S_T	transverse pitch (m)
T	mixed-mean fluid temperature (K or °C)
T_{amb}	ambient temperature (K or °C)
T_b	heat sink base temperature (K or °C)
u	velocity in the finned section (m/s)
u_{app}	approach velocity (m/s)
u_{av}	average velocity along fin surface (m/s)
α	thermal diffusivity (m ² /s)
DT_{lm}	log-mean temperature difference
h_o	overall surface efficiency
m	absolute viscosity (Pa · s)
r	density (kg/m ³)

INTRODUCTION

The correct characterization of pressure loss and heat transfer of heat sinks is critical for their design. In the current work, we compare the heat transfer behavior of a pin fin heat sink of square pin cross section, gathered by experimentation, with a computational model using existing literature as the source for correlations. A similar comparison is made in [1] between measured and calculated pressure loss.

Many authors have undertaken the characterization of parallel plate heat exchangers. Some are reviewed in Dogruoz et al. [1], which investigates the hydraulic characteristics of pin fin heat sinks. A flow network approach was used by Butterbaugh and Kang [2], who detailed the calculation of each network element. They accounted not only for flow bypass, but also for that part of the flow that enters the heat sink and exists through the top, the so-called tip leakage. Using flow information one can find the thermal resistance by the use of correlations for overall thermal resistance or pin averaged heat transfer coefficient.

Flow separation and complex three-dimensional flow at the pin-base junction have kept the work on pin fin heat sinks rather sparse. Simple closed form analytical models of the friction and heat transfer coefficients are not available for either laminar or turbulent regimes. Models, therefore, have been simple one-dimensional formulations with empirical formulations of friction and heat transfer coefficients

[3, 4] or fully conjugate 3-D computational fluid dynamics (CFD) solutions [5, 6].

Except for the pin-end wall effects and tip leakage, the cross flow within the pin-fin heat sink resembles that within an array of tubes of infinite length. Arrays of circular tubes have been extensively studied [7, 8] but detailed experimental data on arrays of tubes of square or rectangular cross-section are rare, except for rectangular cross-sections of large aspect ratio used in in-line and off-set strip fins [4, 7]. Data of this type, although not specified for such small heat exchangers as those within the scope of this paper, have traditionally been used by designers, due to the lack of a better methodology. This is also acknowledged and addressed in this paper.

Shaukatullah et al. [3] measured the thermal performance for in-line square pin fins and plate heat sinks for different fin thickness, spacing, height, and angle of approach for velocities under 5m/s, while allowing flow to partially bypass the exchanger. More recently, Jonsson and Moshfegh [4] experimentally characterized plate and circular, rectangular and strip pin fins, in both staggered and in-line configurations for different dimensions, allowing for variations on tip and side bypass. CFD approaches have been extensively applied to study flow and heat transfer in heat sinks, for example by Jonsson and Moshfegh [5], and Dvinsky et al. [6]. Generally, most of the studies have not been especially useful for extracting local friction or heat transfer coefficients.

This study is motivated by a general lack of data that can be used for heat sink optimization and design and simulation of their thermal behavior. The objectives of this paper are to present carefully documented experimental data on the thermal behavior of a consistent set of square pin fin heat sinks of variable height and pitch, and to compare their behavior to the behavior predicted using data for circular tube arrays of infinite (or very long) length available in the literature. This paper is a companion to Dogruoz et al. [1] where the flows and pressure drop through the same set of heat sink has been previously reported.

EXPERIMENTAL APPARATUS AND MEASUREMENTS

Experimental Apparatus

The heat sinks used in this study are a subset of the set used by Shaukatullah et al. [3] and are summarized in Table 1. The heat sinks are made of aluminum 2024-T6 ($k_{Al}=151 \text{ W/m} \cdot \text{K}$), using electro-discharge-machining. All of the heat sinks have base

dimensions of 25mm x 25mm x 2.5mm where the pin height ranges from 12.5mm to 22.5mm. Experiments were conducted on 4x4, 5x5, 6x6, 7x7 and 8x8 pin arrays for $a=1.5\text{mm}$, and a 5x5 arrangement with $a=2\text{mm}$ and $a=2.5\text{mm}$.

The test geometry is described in Fig.'s 1a and 1b. Further details are available in [1, 9]. The test apparatus was designed to allow tip by-pass variations but not side by-pass. A duct of rectangular cross-section was constructed with 0.5" Plexiglas walls and ceiling, and aluminum floor, where the sides and the floor are fixed and the ceiling is movable. The test section is upstream of a flow management section leading to a small centrifugal blower and runs at sub-ambient pressure, as a way to better controlling flow and velocity conditions. A quarter-circle cross-section flare was placed at the mouth of the channel, as a smooth contraction from the ambient, into the channel. A Pitot tube placed 2.5cm downstream from the entrance (22.5cm upstream from the heat sink) measured the total pressure, which was compared to the first static pressure tap, 18cm from the entrance, to measure approach velocity. During testing, the heat sink was placed on the fixed bottom wall, while the top wall was adjusted to allow for tip by-pass at pre-selected typical clearance ratios. The upstream face of the heat sink was located 30cm from the inlet. The heat sink rests on a 7.5cm x 5cm x 2.5cm Renshape ($k_{rs}=0.13\text{ W/m}\cdot\text{K}$) base. A 2.5cm x 2.5 cm, 70 Ω Kapton-encapsulated Nichrome foil heater is used to provide electrical heating. A 0.6mm thick copper heat spreader was placed between the heater and the heat sink base. The heat sink fits on a seat, such that the floor of the channel is flush with the base of the heat sink, and only the pins protrude. The heat sink was attached to the heater assembly by a thermally conductive adhesive transfer tape. Downstream from the channel, a straw pack was used inside a plenum, to reduce unsteadiness and the effects of flow structures on the test section. To adjust the flow velocity within the test section, an AC controller (Variac) was used on the blower. In addition, a flow by-pass gate in the downstream plenum was used for more delicate flow control. The temperature of the heat sink base was measured using a 36-gauge copper-constantan thermocouple installed in the center of the base through a small hole well drilled into the side of the base. The thermocouple in the heat sink is connected in series with another thermocouple exposed to ambient air, so that the temperature difference is measured directly. A separate ambient temperature junction was referenced to ice to record the absolute ambient temperature. The thermocouple signals were recorded by a Fluke

Hydra 2620 data acquisition unit connected to a personal computer via GPIB. Power was administered to the resistive heater by a 10W in-house power supply delivering constant power. Typical test power was 10W, although it was lower for the cases in which the base temperature exceeded 80°C.

The thermal performance was measured at approach velocities of $u_{app}=2\text{m/s}$ ($\pm 3\%$) and $u_{app}=4\text{m/s}$ ($\pm 1\%$), while measured dynamic head showed an RMS fluctuation of 7% and 1%, respectively. Details of this uncertainty analysis can be found in Dogruoz et al. [1].

In the experiments, it was assumed that the steady state conditions were achieved when the change in the base temperature of heat sink was less than 0.5°C in a time period of 90s. When these conditions were achieved, the base temperature, power delivered to heat sink, and approach velocity were recorded.

Data Reduction

The total thermal resistance, R_{th} is the primary thermal performance metric explored in this study. It is defined as:

$$R_{th} = \frac{T_b - T_{amb}}{q_{conv}} \quad (1)$$

where q_{conv} is the heat convected through the heat sink and given by,

$$q_{conv} = q_{tot} - q_{loss} \quad (2)$$

T_b is the base temperature of the heat sink, T_{amb} is the ambient temperature, q_{tot} is the total amount of power supplied to heat sink, and q_{loss} is the total amount of heat loss due to conduction and radiation.

To estimate the conduction heat losses through the base, a half-symmetry finite element model was constructed utilizing a commercial thermal analysis code. The sides and the bottom of the base module were assumed to be cooled by free convection with a heat transfer coefficient of $3\text{ W/m}^2\cdot\text{K}$. The heat extracted through the top, which were reported as 80% of the total losses, was due to forced convection due to the channel flow. Heat transfer coefficients were estimated by temperature measurements of a flat plate in the place of a heat sink, flushed to the floor as well. The flat plate meets the same specifications as the heat sinks, but with no pins at all, and the measured heat transfer coefficients were approximately $45\text{ W/m}^2\cdot\text{K}$ and $60\text{ W/m}^2\cdot\text{K}$ for approach velocities of 2m/s and 4m/s, respectively, nearly independent of the channel height, H (varied from 12.5mm to 22.5mm). These are the base

parameters used in the finite element model, and by increasing the free convection coefficient to $4 \text{ W/m}^2 \cdot \text{K}$ and that of forced to $80 \text{ W/m}^2 \cdot \text{K}$, an upper bound to the heat losses through these paths was obtained. It was found that the conduction losses were of the order of 10% of the total power.

A transient comparison was made between the cases where pins touched the ceiling, and where a small air gap was present. This comparison showed that tip heat loss by conduction into the ceiling was not very influential.

When estimating energy losses due to radiation, the solution was bounded by a plausible worst and best case scenario. To estimate the upper bound, each pin with an emissivity of 0.8 was allowed to transfer heat directly to the surroundings, which was assumed to be a black body at the ambient temperature. For the lower bound, each pin was assumed to be surrounded by a wall at the same temperature and with the same emissivity as that of the pins. Heat was exchanged to the ambient surroundings through the upper plane only, since the bottom plane is the heat sink floor, assumed to be at the pin temperature as well. In this analysis, the view factor of two co-axial finite-height cylinders was used, while keeping the surface areas of the pin and the wall as fixed.

Uncertainty

The uncertainty in R_{th} , denoted by dR_{th} , will be a combination of parameters that R_{th} depends on and their individual uncertainty [10]. This relationship is defined in Eq. (3), and the individual uncertainties are summarized in Table 2.

$$dR^2 = \left(\frac{1}{q_{tot} - q_{loss}} d(T_b - T_{amb}) \right)^2 + \left(\frac{-(T_b - T_{amb})}{(q_{tot} - q_{loss})^2} dq_{tot} \right)^2 + \left(\frac{T_b - T_{amb}}{(q_{tot} - q_{loss})^2} dq_{loss} \right)^2 \quad (3)$$

To determine dq_{loss} , the loss modes were varied as described above. It is relevant to clarify that the uncertainties found by this approach were not necessarily small, but since q_{loss} is small compared to q_{tot} , the impact of individual uncertainties on dR_{th} is small. The uncertainty in R_{th} is typically between 6% or 8%.

EXPERIMENTAL RESULTS

Thermal resistance was measured for the heat sinks described in Table 1, Fig. 1 for the approach velocities of 2 and 4m/s. Heat sink geometry C3 was evaluated at different velocities to sense its flow velocity dependence on thermal resistance and the

results are shown in Fig. 4. Above 2.4m/s a dependence on approach velocity of a power of 0.73 was found, while below 2.4m/s a trend on u_{app} to the power of 0.42 was observed. It should be noted that the thermal resistance uncertainty for lower velocities tends to be higher and conduction heat losses become more and more comparable to the heat convected. This may skew the data below 2.0m/s.

Figure 5 shows the thermal resistance against non-dimensional pitch (P_L) and height (H) at $u_{app}=4\text{m/s}$. Thermal resistance increases with diminishing height, and the effect is more pronounced for shorter pins. In fact, Metzger et al. [11] reported a Nusselt number from correlations up to 50% higher for infinitely long tubes when experimentally compared with very short ($H/d_f=1$) pins. For shorter pins, the floor becomes an important heat transfer area, with a lower local heat transfer coefficient with respect to the pins. As the pins grow taller, the exposed area in surfaces with higher local heat transfer rates increases, with a compounding effect. As the pitch decreases the thermal resistance decreases due to increased surface area, as the pin count increases, and higher Nusselt numbers [8], as the average fin velocity increases.

Another way to look at the same data is to define a heat transfer coefficient (h) based on log-mean temperature difference (ΔT_{lm}), fin efficiency of unity, and the wetted heat sink area (A_w),

$$h = \frac{q_{conv}}{A_w \Delta T_{lm}} \quad (4)$$

Figure 6 presents the data in this form. Since h is defined on total area, and since the local heat transfer coefficients on the floor are lower than those on the pin [11], variations of pin height cause variations in h due to changes in the flow. Also, variations in h are due to increased importance of the unpinned base area, A_b . We can see that this effect is not so pronounced between 22.5mm and 17.5mm (a decrease of 11% in h , for a 22% decrease in height) as it is between 17.5mm and 12.5mm (a decrease of 31% in h , for a 29% decrease in height). As pitch increases, the floor area to pin area increases (for any given height) and thus h loses value as a pin heat transfer coefficient, making a poor comparison between varying P_L for the same H .

Figure 7 presents the Nusselt number, based on h of Eq. (4), versus the Reynolds number based on the average pin velocity for a pin height of 22.5mm. The data at $H=22.5\text{mm}$ are correlated to within 6% by the expression:

$$Nu = 0.033 Re_{av}^{0.73} P_L^{0.60} \quad (5)$$

Although Nu carries the same limitations as h regarding the loss of validity as P_L increases and the limited span of Re , this correlation serves as a reference for comparison to classical correlations for tube bundles.

Observing the influence of top clearance ratio on the thermal resistance, shown in Fig. 8, R_{th} has a relatively low sensitivity to CL for a given P_L . This is due to the low sensitivity of R_{th} to u_{app} as seen in Fig. 4. From [1], we know that a heat sink under by-pass flow will have lower flow rates going through the array, and if the equivalent flow rate is imposed, the same heat sink without by-pass will present the same pressure drop. As by-pass is allowed, the flow through the array diminishes, leading to the mild increases observed in R_{th} .

Figure 9 presents the variation of R_{th} vs. P_L at various clearance ratios. The sparser heat sinks exhibit greater clearance ratio dependence than the denser heat sinks even though the by-pass ratio is higher in the case of the latter [1]. If $CL \neq 0$, as the pitch continues to decrease, a minimum is expected, where the increasing wetted area is not enough to balance the decreasing heat transfer rates due to diminishing flow through the array. Jubran et al. [12] found the optimum to occur at $P_L=2.5$, for circular pins, interacting floor and $H/d_f=9.4$. Although this optimal pitch was not achieved with the heat sinks tested, Figure 21 shows the predicted values of R_{th} vs. P_L from the computational model. Further discussion will be given as a case study at the end of this paper.

In Fig. 10, the thermal resistance at different clearance ratios has been normalized by the corresponding zero clearance thermal resistance, $(R_{th})_0$. We find that the increase of relative thermal resistance due to clearance ratio lie within a $\pm 4\%$ band (well within uncertainty, estimated at about 6%-8%) about the linear fit. The increase in thermal resistance from $CL=0$ to $CL \rightarrow \infty$ nears 25% for all heat sinks.

COMPUTATIONAL ANALYSIS

A simple one-dimensional model employing empirical friction factor correlations was developed by Dogruoz et al. [1] to determine pressure drop in cross-flow pin-fin heat sinks. The so-called “two-branch by-pass model” neglects tip leakage from the finned section to the by-pass channel. Hence, by the continuity equation, each section has a constant mass flow rate of air in the flow direction. Velocities in both of the channels were assumed to be uniform and the momentum equations for a given control volume were written in terms of average velocities. As

explained in [1], average velocity and pressure at a given location could be determined with an iterative procedure for any by-pass clearance provided that the exit static pressures were same the for both ducts. It was generally found that the inlet pressure drop is quite severe and hence input loss coefficients must be accurate for correct predictions. The pressure drop through the core was predicted to within 15% by the use of published friction loss coefficients for circular tubes [8].

In order to design and optimize fin heat sinks, it is necessary to predict the heat transfer by the combined conduction through the pins and convection from their surfaces. Experimentally, it is difficult to measure pin by pin heat transfer coefficients and far easier to measure overall base to ambient thermal resistance. The one-dimensional model requires the specification of a local pin-averaged heat transfer coefficient; hence, the model can be used to test the validity of correlations for the specified heat transfer coefficient by comparing measured and predicted overall thermal resistances. This approach is illustrated in the following section.

The computational algorithm is diagrammed in Fig. 11. The flow problem is first solved to determine the average velocity in the finned section. Since the tip leakage is neglected, the energy balance is written for CV-1 in Fig. 12 as:

$$q_{f,r} + \dot{m}c_p T_x = \dot{m}c_p T_{x+\Delta x} \quad (6)$$

where $\dot{m}$ is the mass flow rate in the finned section, c_p is the specific heat of the fluid, and $q_{f,r}$ is the rate of heat transfer per unit time per row. If the outlet mixed mean fluid temperature is written in terms of its Taylor series with second and higher order terms neglected, the increase in mixed mean temperature of the fluid per row can be determined from,

$$\Delta T = \frac{q_{f,r}}{\dot{m}c_p} \quad (7)$$

and the total heat transfer rate convected through the heat sink can be written as,

$$q_{conv} = \sum_{i=1}^{n_r} (q_{f,r})_i \quad (8)$$

where n_r is the number of fin rows.

If the base temperature of the heat sink, T_b is averaged and assumed to be constant, the total convective heat transfer rate from the heat sink to the fluid per row can be written as,

$$q_{f,r} = h_o h (n_f A_f + A_b) (T_b - T) \quad (9)$$

where h_o is the overall surface efficiency, n_f is the number of fins per row, A_f is the exposed surface area

of a single fin, A_b is the corresponding base surface area for a given control volume, and T is the local mixed mean air temperature. For rectangular straight or pin fins, h_o can be expressed as a function of areas subjected to heat transfer and the corresponding fin efficiency,

$$h_o = 1 - \frac{n_f A_f}{(n_f A_f + A_b)} \left(1 - \frac{\tanh(m L_c)}{m L_c} \right) \quad (10)$$

where m is the fin parameter,

$$m = \sqrt{\frac{h P_f}{k A_{cs}}} \quad (11)$$

and L_c is the effective fin length given as,

$$L_c = H + \frac{d_f}{4} \quad (12-a)$$

for rectangular pin fins,

$$L_c = H + \frac{a}{2} \quad (12-b)$$

for straight pin fins. P_f and A_{cs} are the perimeter and the cross-sectional area of a single fin, respectively.

For closely spaced infinitely long circular tube bundles, the empirical Nu correlation proposed by Zhukauskas [2] appears to be a good first order approximation for square pin fins, in lieu of measuring the values directly. Its form is:

$$Nu = C Re_{av}^n Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)^{1/4} \quad (13)$$

where Re_{av} can be written as,

$$Re_{av} = \frac{\rho u_{av} d_f}{\mu} \quad (14)$$

In Eq. (13), Nu is the Nusselt number, C and n are constants depending on the type of fin arrangement, and Re_{av} , Pr and Pr_s are the Prandtl numbers of the fluid at the mixed mean and surface temperatures, respectively. In Eq. (14), ρ is the density, μ is the absolute viscosity and u_{av} is the mean velocity along the fin surface. Determining Nu from Eq. (13), heat transfer coefficient can be written as,

$$h = \frac{Nu}{d_f} k \quad (15)$$

where k is the thermal conductivity of the fluid.

RESULTS AND DISCUSSION

Although it is not possible to compare local behavior, comparison of the predicted overall thermal resistance with the data is a useful exercise in determining the accuracy of the one-dimensional model. Figure 13 presents the experimental thermal resistance results and corresponding computational predictions of the heat sink with $P_L=3.133$ in the approach velocity range of $0.8\text{m/s} \leq u_{app} \leq 5.3\text{m/s}$. At

low approach velocities, the discrepancy between the theoretical and the experimental results reaches around 34%. However, at higher approach velocities, $u_{app} \geq 3.0\text{m/s}$, there is good agreement. It is worth noting that a similar behavior was also observed in the case of hydraulics resistance [1]. For this particular case, the power dependence of Nu on Re_{av} , thus that of h on u_{app} is about 0.5 for the empirical correlations [8]. For the experiments, this power-law exponent is 0.42 for the approach velocities less than 2.4m/s and 0.73 for the higher velocities. This discrepancy suggests a possible failure of Eq. (13) in both of the laminar and turbulent regimes, perhaps due to differences between circular and square tube behavior. Figures 14 and 15 demonstrate the thermal resistance of all of the tallest heat sinks ($H=22.5\text{mm}$) with $a=1.5\text{mm}$ and approach velocities of 2 and 4m/s . Again, good agreement is found at higher approach velocities with poor agreement at lower approach velocities.

In order to explore the effect of heat sink height, H , experiments and computations were performed for three different cases of $H=12.5$, 17.5 and 22.5mm at $u_{app}=2$, 4m/s and the corresponding results are shown as R_{th} vs. P_L in Fig.'s 16, 17 and 18, respectively. Taller pins not only have a larger surface area, but also have a higher heat transfer coefficient as a result of diminishing end-wall effects and pin-base interaction. At higher approach velocities, computational results match well with those of experiments, surprisingly, even for the shortest heat sinks. Decreasing the heat sink height, the discrepancy between the experimental results and theoretical predictions increases, with the worst occurring for the shortest heat sink with $u_{app}=2\text{m/s}$, at the largest pin spacing. It is hypothesized that end wall effects are greatest for the shortest heat sinks; hence departure from Eq. (13) is the greatest in that case, at low velocities. These plots also demonstrate a monotonic trend of increasing thermal resistance with increasing pitch, which can be attributed to a decrease of surface area at higher non-dimensional pitch sizes for an equal size base area. As the pitch, or pin spacing increases, the exposed base area, A_b , becomes a larger percentage of the total area. Hence, the assumption that the heat transfer is the same on the pins and on the base may become poorer at larger pin spacing.

The one-dimensional model can also be compared to data in the cases with tip clearance. It should be remembered that tip leakage is not accounted for this model and may inherently introduce error. Figures 19 and 20 demonstrate the thermal resistance against the clearance ratio at $P_L=2.238$ (the densest

configuration) and $P_L=3.917$ (the sparsest configuration), respectively. For the densest configuration, there is an excellent agreement between the theory and the experiments in the case of $u_{app}=4\text{m/s}$ especially at high clearance ratios. The agreement is even better for $P_L=3.917$. However, the discrepancy between the results gets larger if u_{app} is decreased to 2m/s , where the worst case is obtained at zero clearance and the error between the results is up to nearly 32% at $P_L=2.238$. While not shown here, a similar behavior can also be seen for non-dimensional pitch sizes of $P_L=2.611$ and 3.133 . Regardless of the pitch and the approach velocity, both of the theoretical predictions and the experimental data follow the same trend with the increasing clearance ratio.

Case Study: Heat sink optimization

In designing electronic packages, major constraints are the highest temperature and the geometric features of the package. Due to a limited space in many applications, optimization of the heat sinks is one of the major issues to be taken into consideration in electronic equipment design. Although the main goal of this paper is to investigate the applicability of infinitely long tube correlations to compact heat exchangers, a case study was performed for the optimization of heat sinks utilized in the experimental study, since this is the final step in the electronics cooling if the design constraints mentioned above are specified.

In this case study, it is intended to determine the optimum number of fins for a fixed size of heat sink base ($25\text{mm}\times 25\text{mm}$), height ($H=22.5\text{mm}$) and a fin diameter ($d_f=1.5\text{mm}$). In addition, the base temperature (height), T_b is assumed to be constant and the approach velocity is taken to be $u_{app}=4\text{m/s}$. For the given conditions, a simple optimization can be performed by analyzing the variation of thermal resistance with respect to pitch size, or number of fins. Figure 21 illustrates this case at various clearance ratios. For the clearance ratios of 0.5, 1.0 and 2.0, the minimum thermal resistance is obtained at $P_L=1.96$, where the corresponding number of fins is 9. This is lower than that of predicted in [12]. Flow by-pass increases with decreasing pitch [1], therefore further reduction of the pitch does not decrease the thermal resistance as the fin velocity decreases with increasing by-pass ratio.

CONCLUSIONS

Thermal resistance and performance results were shown for in-line pin square pin fin heat sinks with various pitch sizes. A computational model was derived and the model predictions were compared

with the experimental data. A case study was performed to demonstrate the fin optimization problem for certain design constraints.

Based on the results of the present study, the following conclusions can be drawn:

1. Overall thermal resistances for fixed pin height and zero by-pass suggest laminar behavior below approach velocities of 2.4m/s , and turbulent behavior above 4m/s .
2. Variations in overall thermal resistances due to variation of pin height and pitch are not well correlated solely to variations in total area, suggesting that there are important differences between the heat transfer coefficients on the pins and on the base area.
3. For the tallest pins, the overall thermal resistance increases by no more than 25% from its zero clearance value to its value at a clearance ratio of 3.0.
4. The case of conventional heat transfer correlation for circular tube bundles was only moderately successful for predicting pin fin heat sink performance. The model significantly underestimated the thermal resistance at low velocities and was in good agreement at higher velocities.
5. The two-leg by-pass model demonstrated excellent ability to predict the effects of flow by-pass on the overall thermal resistance, in the turbulent flow regime, suggesting that the heat transfer from the heat sink surfaces is set almost completely by the local fin velocity. Apparently, tip leakage effects are minor.
6. The optimal pin spacing that minimizes the overall thermal resistance, at fixed approach velocity, is predicted to be $P_L\approx 2.0$ by the model. This was not experimentally verified.

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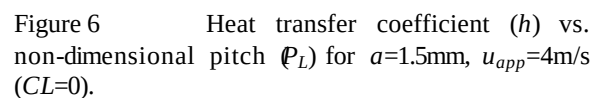
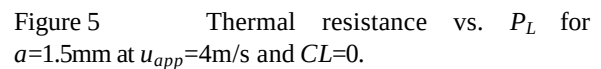
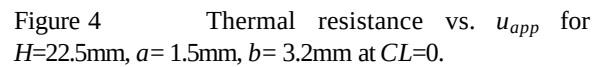
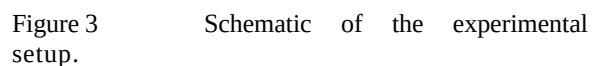
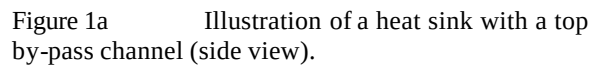
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Table 1 Geometric description of heat sinks tested. All dimensions in mm. n : number of fins in a row or column.

Desc.	a	b	H	n	$P_L=(a+b)/a$
1A	1.50	1.86	12.50	8	2.238
1B	1.50	1.86	17.50	8	2.238
1C	1.50	1.86	22.50	8	2.238
2A	1.50	2.42	12.50	7	2.611
2B	1.50	2.42	17.50	7	2.611
2C	1.50	2.42	22.50	7	2.611
3A	1.50	3.20	12.50	6	3.133
3B	1.50	3.20	17.50	6	3.133
3C	1.50	3.20	22.50	6	3.133
4A	1.50	4.38	12.50	5	3.917
4B	1.50	4.38	17.50	5	3.917
4C	1.50	4.38	22.50	5	3.917
5A	1.50	6.33	12.50	4	5.222
5B	1.50	6.33	17.50	4	5.222
6A	2.00	3.75	12.50	5	2.875
6B	2.00	3.75	17.50	5	2.875
6C	2.00	3.75	22.50	5	2.875
7A	2.50	3.13	12.50	5	2.253
7B	2.50	3.13	17.50	5	2.253
7C	2.50	3.13	22.50	5	2.253

Table 2 Uncertainties on R_{th} and its parameters.

$d(T_b - T_{amb})$	dq_{tot}/q_{tot}	dq_{loss}/q_{loss}	dR_{th}/R_{th}
$\approx 0.5^\circ\text{C}$	0.5%	20%	6%-8%



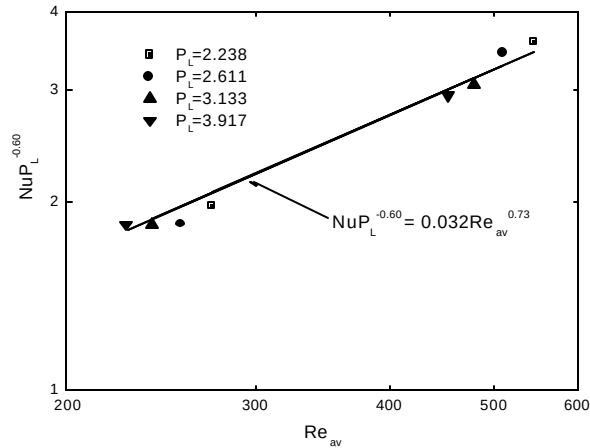


Figure 7 Nusselt number vs. Reynolds number (based on average velocity, u_{av}) for $CL=0$, $a=1.5\text{mm}$, $H=22.5\text{mm}$ under $u_{app}=2\text{m/s}$ and 4m/s .

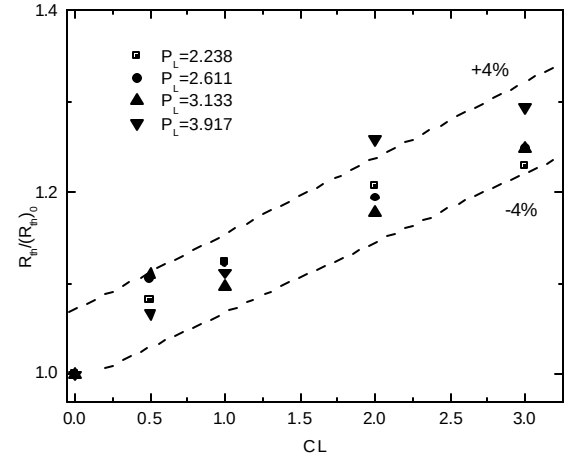


Figure 10 Thermal resistance (R_{th}) normalized with $(R_{th})_0$ vs. top clearance for $a=1.5\text{mm}$, $u_{app}=4\text{m/s}$ at $H=22.5\text{mm}$ ($CL=0$).

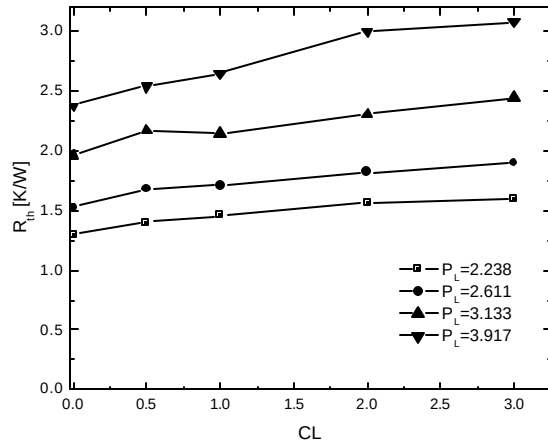


Figure 8 Thermal resistance vs. clearance ratio for $H=22.5\text{mm}$, $a=1.5\text{mm}$ at $u_{app}=4\text{m/s}$.

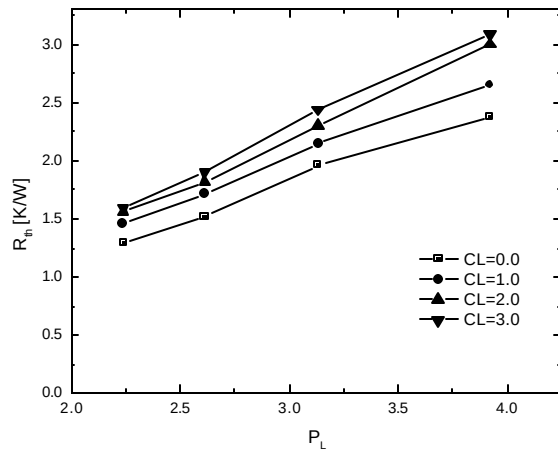


Figure 9 Thermal resistance vs. P_L for various CL , $a=1.5\text{mm}$, $H=22.5\text{mm}$ at $u_{app}=4\text{m/s}$.

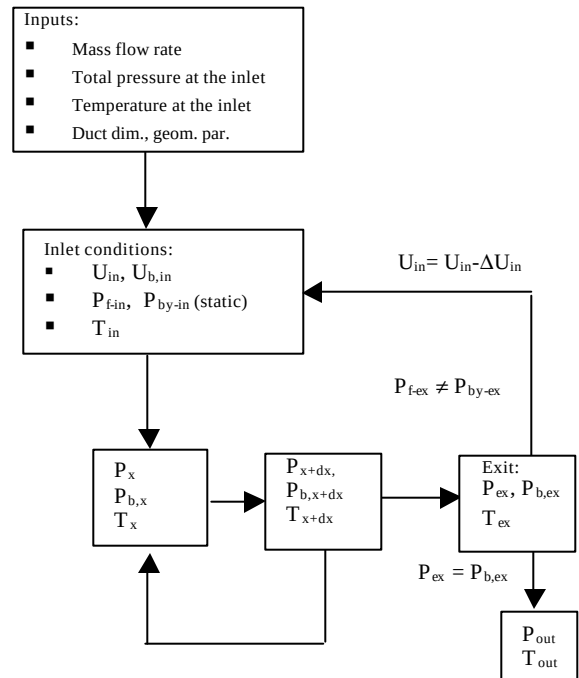


Figure11 Block diagram demonstrating the computational procedure to determine pressure drop and heat transfer rate in cross-flow heat sinks.

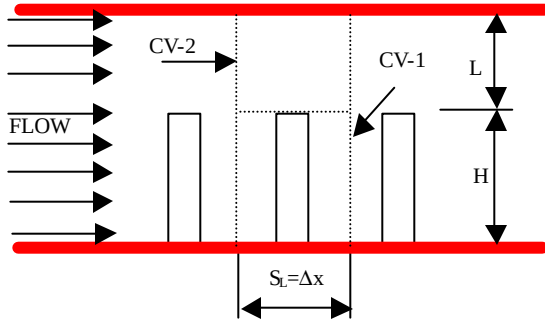


Figure 12 Control volumes for finned and bypass sections in the theoretical model.

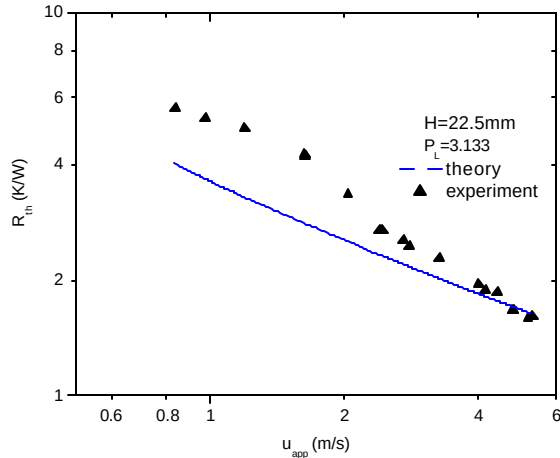


Figure 13 Comparison of predicted and experimental thermal resistance (R_{th}) vs. u_{app} at $CL=0$, $P_L=3.133$ and $H=22.5\text{mm}$.

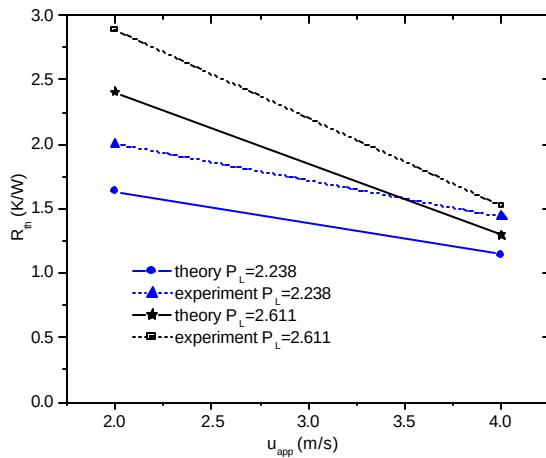


Figure 14 Comparison of predicted and experimental thermal resistance (R_{th}) vs. u_{app} at $CL=0$, $H=22.5\text{mm}$ for $P_L=2.238$ and 2.611 .

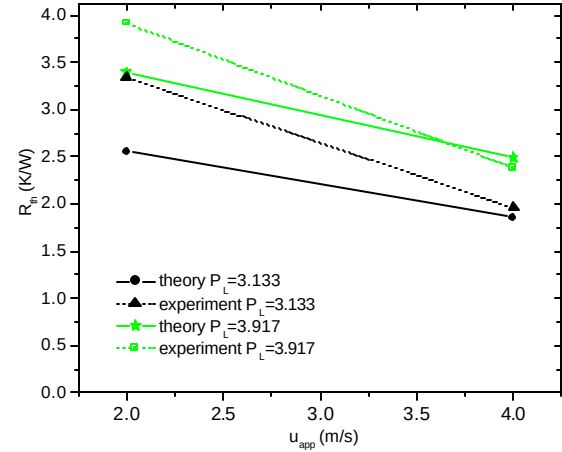


Figure 15 Comparison of predicted and experimental thermal resistance (R_{th}) vs. u_{app} at $CL=0$, $H=22.5\text{mm}$ for $P_L=3.133$ and 3.917 .

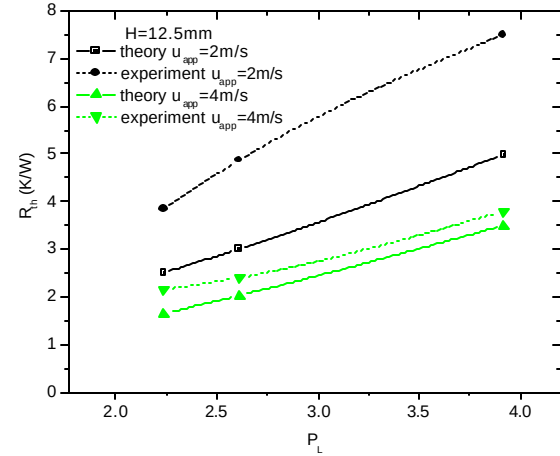


Figure 16 Comparison of predicted and experimental thermal resistance (R_{th}) vs. non-dimensional pitch (P_L) at $CL=0$ and $H=12.5\text{mm}$ under $u_{app}=2\text{m/s}$ and 4m/s .

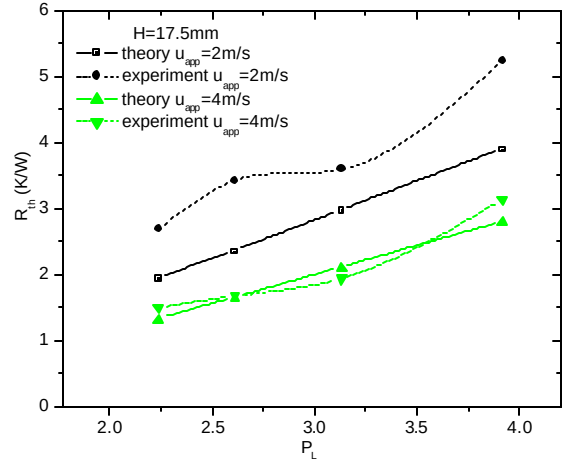


Figure 17 Comparison of predicted and experimental thermal resistance (R_{th}) vs. non-dimensional pitch (P_L) at $CL=0$ and $H=17.5\text{mm}$ under $u_{app}=2\text{m/s}$ and 4m/s .

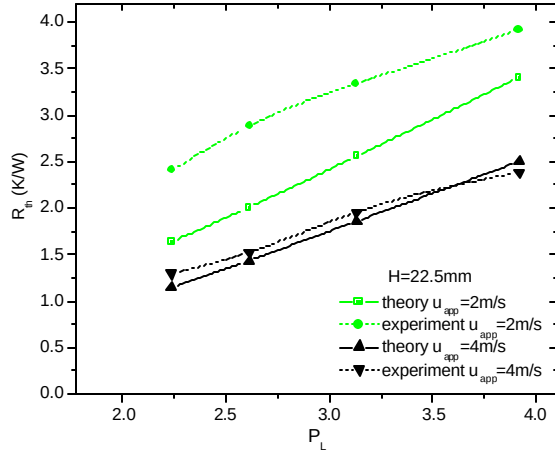


Figure 18 Comparison of predicted and experimental thermal resistance vs. P_L at $CL=0$, $H=22.5\text{mm}$ under $u_{app}=2\text{m/s}$ and 4m/s .

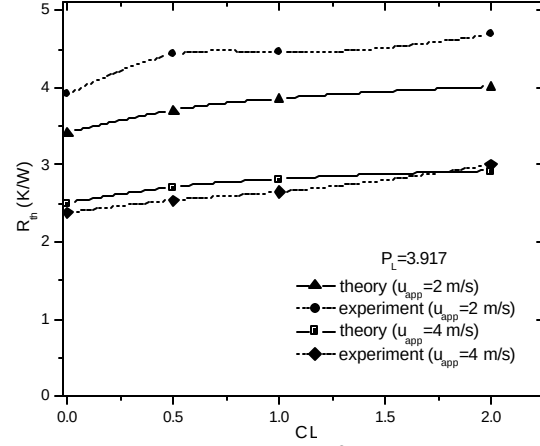


Figure 20 Comparison of predicted and experimental thermal resistance vs. CL at $P_L=3.917$, $H=22.5\text{mm}$ under $u_{app}=2\text{m/s}$ and 4m/s .

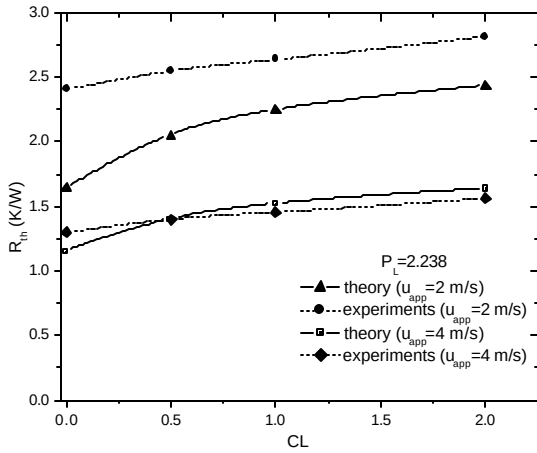


Figure 19 Comparison of predicted and experimental thermal resistance vs. CL at $P_L=2.238$, $H=22.5\text{mm}$ under $u_{app}=2\text{m/s}$ and 4m/s .

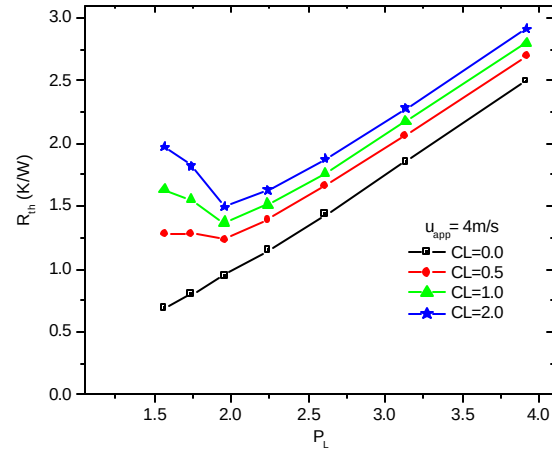


Figure 21 Thermal resistance vs. P_L at various clearance ratios, $H=22.5\text{mm}$ and $u_{app}=4\text{m/s}$.